Statistics for the Overall Findings Storyboard (overall_v1)

Colter Bay, Grand Teton National Park, 2022

Nigel K. Anderson · American Museum of Natural History

2026-11-04

All percentages are back-transformed from the log scale: % change in detections per km away from the parking lot, or % difference under white vs. red light. 95% CIs are Wald intervals; p-values are Wald z-tests. "Holm" and "BH" mark p-values adjusted across the five intensities (Holm) or across the taxa tested (Benjamini–Hochberg). Light color and intensity were set experimentally; distance is observational.

Model formulas

All models are fitted with `glmmTMB` (negative binomial, `nbinom2`). `site` and `date` (night) are random intercepts; light `color` (R/W) and `intensity` (10, 30, 50, 70, 100%) are factors; `dist_km` = distance from the parking lot; `jd_c` = day of year, centered.

Figures 1B–C, 2 and 5B–C — one model per taxon (Figures 1–2: 11 stations; Figure 5: 13 stations, GRTE15 excluded). Figure 1 uses the color × distance slopes; Figure 2 uses the intensity × distance slopes (A–B) and the color × intensity contrasts (C–D).

Bats (total detections per site-night; no zero-inflation):

```
total_det ~ color:intensity + intensity:dist_km + color:dist_km + color + intensity
  + dist_km + mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2)
  + (1 | site) + (1 | date)
```

Moths (specimens per trap night; no zero-inflation):

```
detections ~ color:intensity + intensity:dist_km + color:dist_km + color + intensity
  + dist_km + mean_phase + pct_nonforest + jd_c + I(jd_c^2) + brightness_dark
  + (1 | site) + (1 | date)
```

Figure 3 — bat hierarchy (zero-inflated, `ziformula = ~1`; zero-filled site-nights). All bats uses total detections; guilds and species use their own detections with `taxon` = guild or species. Every tier converged with the full covariate set.

```
All bats:          total_det ~ dist_km + color + intensity + mean_phase + pct_nonforest
                     + brightness_dark + jd_c + I(jd_c^2) + (1 | site) + (1 | date)
Guilds / species:  detections ~ taxon * dist_km + color + intensity + mean_phase
                     + pct_nonforest + brightness_dark + jd_c + I(jd_c^2) + (1 | site) + (1 | date)
```

Figure 4 — moth hierarchy (zero-inflated, `ziformula = ~1`; one model per taxonomic level over the taxa with at least 30 site-nights, zero-filled trap nights). If a level's model did not converge, covariates were dropped in this order: non-forest cover, sky brightness, season^2, moon phase.

```
detections ~ taxon * dist_km + color + intensity + mean_phase + pct_nonforest
  + brightness_dark + jd_c + I(jd_c^2) + (1 | site) + (1 | date)
```

In both hierarchies a node's slope and p-value come from the taxon-specific `dist_km` slope (`emtrends`); curves are averaged over color and intensity.

Figure 1A and 5A: Station sky brightness

Table 1: Sky brightness from the fisheye photos (dark sky units). Lights off = darkest lights-off photo (as used in the models); white 100% = mean of three photos (GRTE11 p1 excluded: transient light). Distance from the reference point at GRTE17.

Station	Distance (km)	Lights off (DSU)	White 100% (DSU)	Mapped (DSU)	In Figure 1
GRTE17 (lot)	0.00	2.9	28.6	28.6	yes
GRTE16 (lot)	0.37	15.3	83.2	83.2	yes
GRTE05	0.49	8.4	11.9	11.9	yes
GRTE01 (lot)	0.51	5.9	20.1	20.1	yes
GRTE06	0.77	3.1	3.6	3.6	yes
GRTE02	0.98	4.1	3.5	3.5	yes
GRTE08	1.56	4.3	2.2	2.2	yes
GRTE03	1.60	3.7	3.9	3.9	yes
GRTE09	1.92	4.0	—	4.0*	yes
GRTE11	2.02	4.3	4.9	4.9	yes
GRTE10	2.10	2.7	—	2.7*	yes
GRTE15	7.20	3.3	3.6	3.6	no (far)
GRTE14	7.40	24.0	9.6	9.6	no (far)
GRTE13	7.46	2.8	3.0	3.0	no (far)

* No white-light photos; the lights-off value is mapped instead. GRTE15 appears in neither map. The surface in the maps is inverse-distance-weighted interpolation of log DSU (power 2), clipped to 300 m around the stations.

Figure 1B–C: Distance slopes by light color (11 stations, 0–2.1 km)

Table 2: Models (negative binomial GLMMs; random intercepts for site and night).

Model	Obs.	Stations	Family	Converged
Bats: total detections per site-night	502	11	nbinom2	yes
Moths: specimens per trap night	176	11	nbinom2	yes

Both models: color × intensity, intensity × distance, color × distance, moon phase, non-forest cover, sky brightness (lights off), season (day of year, linear + quadratic), and random intercepts for site and night (date). The night effect makes nights, not site-nights, the replicates for the light treatments, which were set night by night. Full formulas: see *Model formulas* above.

Table 3: Figure 1B — bats: % change in detections per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	-56%	-79 to -9%	-2.20	0.028
White	-54%	-78 to -6%	-2.12	0.034

Table 4: Figure 1C — moths: % change in specimens per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	+4%	-21 to +38%	0.30	0.761
White	+62%	+24 to +112%	3.55	< 0.001

Figure 2: Effects at each light intensity (same models as Figure 1)

Table 5: Figure 2A — bats: % change in detections per km away from the lot, by intensity.

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	-55%	-78 to -6%	-2.13	0.033	0.099
30%	-44%	-73 to +18%	-1.52	0.128	0.128
50%	-55%	-78 to -4%	-2.08	0.038	0.099
70%	-60%	-81 to -16%	-2.43	0.015	0.068
100%	-60%	-81 to -17%	-2.47	0.014	0.068

Table 6: Figure 2B — moths: % change in specimens per km away from the lot, by intensity.

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	-11%	-42 to +37%	-0.52	0.600	1.000
30%	+24%	-18 to +87%	1.01	0.312	0.936
50%	+84%	+24 to +173%	3.02	0.003	0.010
70%	+69%	+22 to +136%	3.13	0.002	0.009
100%	+9%	-26 to +60%	0.43	0.665	1.000

Table 7: Figure 2C — bats: % difference under white vs. red light, by intensity (at the mean distance).

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	+10%	-29 to +69%	0.42	0.677	0.677
30%	-54%	-74 to -18%	-2.63	0.009	0.043
50%	-40%	-62 to -4%	-2.15	0.032	0.126
70%	-35%	-58 to +0%	-1.96	0.050	0.151
100%	+52%	-4 to +139%	1.80	0.072	0.151

Table 8: Figure 2D — moths: % difference under white vs. red light, by intensity (at the mean distance).

Intensity	Estimate	95% CI	z	p	p (Holm)
10%	-4%	-52 to +92%	-0.12	0.904	1.000
30%	-37%	-68 to +26%	-1.31	0.191	0.764
50%	-22%	-57 to +44%	-0.78	0.433	1.000
70%	+13%	-32 to +89%	0.48	0.631	1.000
100%	+154%	+45 to +344%	3.27	0.001	0.005

Red-light nights at 100% intensity: moth traps 4 nights (12–13 July, 5–6 August 2022), bat detectors 6 nights (11–13 July, 4–6 August); they include the July full moon and the two wettest nights of the study.

Figure 3: Bat distance hierarchy (11 stations)

Table 9: Figure 3 — distance slope at each node. All bats: single test (unadjusted); guilds and species: Benjamini–Hochberg across the 12 nodes.

Tier	Taxon	% per km	95% CI	z	p (raw)	p (adj.)
All bats	All bats	-56%	-79 to -10%	-2.23	0.026	0.026
Guild	Low frequency	-32%	-70 to +53%	-0.94	0.348	0.417
Guild	Myev	-47%	-76 to +21%	-1.51	0.132	0.198
Guild	High frequency	-62%	-83 to -14%	-2.32	0.021	0.062
Species	Epfu	-79%	-92 to -48%	-3.36	< 0.001	0.005
Species	Laci	+12%	-55 to +182%	0.25	0.805	0.805
Species	Lano	+16%	-54 to +192%	0.32	0.746	0.805
Species	Myca	-81%	-93 to -47%	-3.21	0.001	0.005
Species	Myci	-57%	-83 to +9%	-1.78	0.076	0.152
Species	Myev	-47%	-79 to +33%	-1.35	0.176	0.235
Species	Mylu	-61%	-84 to -2%	-2.00	0.046	0.110
Species	Myvo	-79%	-92 to -47%	-3.31	< 0.001	0.005
Species	Myyu	-54%	-82 to +20%	-1.59	0.111	0.191

Zero-inflated negative binomial GLMMs of detections per site-night (all bats = total; guild and species = their detections, zero-filled) with taxon × distance, light color, intensity, moon phase, non-forest cover, sky brightness, season, and random intercepts for site and night. Covariates were dropped in order if a model did not converge; covariates used: all bats — mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2); guilds — mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2); species — mean_phase + pct_nonforest + brightness_dark + jd_c + I(jd_c^2).

Figure 4: Moth distance hierarchy (11 stations)

Table 10: Figure 4 — the 18 taxa shown: Benjamini–Hochberg-adjusted $p < 0.05$ across all 51 taxa tested.

Level	Taxon	% per km	p (raw)	p (BH)
Family	Erebidae	+115%	< 0.001	< 0.001
Family	Lasiocampidae	+94%	< 0.001	0.002
Family	Sphingidae	+102%	< 0.001	0.002
Subfamily	Arctiinae	+117%	< 0.001	< 0.001
Subfamily	Smerinthinae	+159%	< 0.001	< 0.001
Subfamily	Sterrhinae	+164%	< 0.001	< 0.001
Subfamily	Lasiocampinae	+83%	< 0.001	0.004
Genus	Smerinthus	+202%	< 0.001	< 0.001
Genus	Apantesis	+167%	< 0.001	< 0.001
Genus	Scopula	+177%	< 0.001	0.002
Genus	Malacosoma	+126%	0.001	0.005
Genus	Feltia	+143%	0.003	0.009
Genus	Xestia	+158%	0.006	0.018
Genus	Euxoa	+95%	0.007	0.022
Species	Smerinthus ophthalmica	+201%	< 0.001	0.002
Species	Apamea inficita	+205%	< 0.001	0.002
Species	Apantesis williamsii	+164%	0.002	0.006
Species	Malacosoma californica	+127%	0.009	0.026

Taxon × distance zero-inflated negative binomial GLMM at each taxonomic level (taxa with at least 30 site-nights), with light color, intensity, moon phase, non-forest cover, sky brightness, season, and random intercepts for site and night.

Table 11: All other taxa tested (not significant after adjustment).

Level	Taxon	% per km	p (raw)	p (BH)
Family	Notodontidae	+63%	0.024	0.058
Family	Noctuidae	+39%	0.049	0.103
Family	Geometridae	+13%	0.480	0.544
Subfamily	Noctuinae	+35%	0.071	0.139
Subfamily	Sphinginae	-44%	0.105	0.191
Subfamily	Erebinae	+41%	0.133	0.219
Subfamily	Pantheinae	-30%	0.249	0.343
Subfamily	Ennominae	-17%	0.286	0.365
Genus	Leucania	+82%	0.019	0.052
Genus	Apamea	+72%	0.021	0.054
Genus	Spilosoma	+98%	0.029	0.068
Genus	Drasteria	+82%	0.033	0.073
Genus	Egira	+45%	0.101	0.191
Genus	Agrotis	+47%	0.131	0.219
Genus	Arctia	-37%	0.209	0.324
Genus	Pero	-31%	0.205	0.324
Genus	Polia	+41%	0.230	0.339
Genus	Sphinx	-37%	0.239	0.339
Genus	Orthosia	+36%	0.271	0.354
Genus	Trichordestra	-21%	0.354	0.430
Genus	Panthea	-17%	0.597	0.621
Genus	Macaria	+6%	0.800	0.800
Species	Spilosoma vagans	+92%	0.062	0.126
Species	Egira curialis	+50%	0.130	0.219
Species	Sphinx vashti	-40%	0.237	0.339
Species	Macaria adonis	+40%	0.257	0.344
Species	Agrotis vancouverensis	+36%	0.343	0.427
Species	Trichordestra liquida	-21%	0.447	0.531
Species	Apamea devastator	+27%	0.459	0.532
Species	Pero behrensaria	-22%	0.498	0.552
Species	Leucania commoides	+23%	0.539	0.585
Species	Panthea virginarius	-21%	0.551	0.585
Species	Apamea unanimis	+17%	0.609	0.621

Figure 5B–C: Distance slopes with the far stations (13 stations, 0–7.5 km; GRTE15 excluded)

Table 12: Models (negative binomial GLMMs; random intercepts for site and night).

Model	Obs.	Stations	Family	Converged
Bats: total detections per site-night	593	13	nbinom2	yes
Moths: specimens per trap night	210	13	nbinom2	yes

Table 13: Figure 5B — bats: % change in detections per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	-1%	-20 to +22%	-0.10	0.922
White	-2%	-20 to +22%	-0.15	0.881

Table 14: Figure 5C — moths: % change in specimens per km under each light color (not adjusted).

Light	% per km	95% CI	z	p
Red	+4%	-4 to +12%	0.95	0.343
White	+17%	+9 to +25%	4.37	< 0.001

Same model structure as Figure 1B–C, fitted on the 13 stations.